A

CONSULTANCY REPORT

On

AI CareerPilot

AI-Powered Career and Skill Recommendation System

submitted in partial fulfillment of the requirement for the award of the degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE & ENGINEERING

Client Organization:

Zenith TEKSystems LLP

Duration:

Jan 2026 – April 2026

Consultancy Amount:

₹ 50,000

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(UGC Autonomous)

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Accredited with 'A' grade by NAAC | Accredited by NBA

Department of Computer Science & Engineering

CERTIFICATE

This is to certify that the consultancy project titled "AI CareerPilot**:** AI-Powered Career and Skill Recommendation System" has been successfully carried out by **Mohammed Fawzaan (160922733167)**, **Syed Towhid Hussain Magrabi (160922733159), Abu Sufyan (160922733175)** and **Syed Hasnain Tariq (160922733150)** from the Department of Computer Science & Engineering, Lords Institute of Engineering & Technology, in partial fulfillment of consultancy work for lords institute of engineering & technology, during the academic year 2025–2026.

The project work has been completed under the guidance and supervision of **Dr. Syed Raziuddin**, and it is certified that this work is original and has not been submitted elsewhere for any other academic or professional requirement.The consultancy objectives were successfully achieved, and all deliverables have been submitted to the client organization.This is further certified that the work done under my guidance, and the results of this work have not been submitted elsewhere for the award of any of the degree

Head of the Department

Dr. Syed Raziuddin
HOD - CSE

Principal

Dr. Ravi Kishore Singh

Date: _____

Place: Hyderabad

EXECUTIVE SUMMARY

The consultancy project titled **"AI CareerPilot: AI-Based Career and Skill Recommendation System"** was undertaken with the objective of designing and developing an intelligent, scalable, and user-centric platform that delivers personalized career guidance using Artificial Intelligence. The project was initiated to overcome the limitations of traditional career counseling approaches, which are largely manual, time-consuming, and lack customization for individual users. The execution began with a comprehensive requirement analysis phase, involving detailed interactions with stakeholders to understand the challenges faced in career planning and guidance. Key requirements such as career assessment, skill gap identification, resume and cover letter generation, interview preparation, and job recommendations were identified and documented. Based on these insights, a robust and modular system architecture was designed, integrating frontend, backend, and AI components along with a secure and efficient database system to ensure seamless data flow and user interaction.

During the development phase, modern technologies such as Next.js for frontend development, Node.js for backend services, and PostgreSQL for database management were utilized to build a responsive and scalable web application. Advanced Artificial Intelligence techniques, including Natural Language Processing (NLP) and Large Language Models (LLMs), were integrated to enable intelligent decision-making and personalized recommendations. The system was developed with multiple functional modules such as an AI-driven career recommendation engine, automated resume and cover letter generator, skill gap analysis module, mock interview simulation, and job search and recommendation system. These modules work in coordination to provide real-time, data-driven outputs tailored to user profiles, ensuring accuracy and relevance. Additionally, emphasis was placed on system performance, scalability, data privacy, and security to ensure a reliable and efficient user experience across different scenarios and user loads.

The system was subjected to rigorous testing processes, including unit testing, integration testing, and user acceptance testing, to validate its functionality, performance, and reliability.

INTRODUCTION

In today's rapidly evolving digital landscape, organizations are increasingly adopting advanced technologies to improve operational efficiency, decision-making, and overall business performance. One of the key technologies enabling enterprise-level transformation is SAP (Systems, Applications, and Products), which integrates various business processes into a unified system. Companies offering SAP technical development services play a crucial role in helping organizations implement, customize, and optimize these enterprise solutions. Zenith TEKSystems LLP is one such emerging organization that specializes in providing customized SAP technical development services and end-to-end support for enterprise applications.

Zenith TEKSystems LLP focuses on delivering industry-tailored SAP solutions, including application management, system integration, migration, and technical consulting. The company emphasizes a customer-centric approach, agile development practices, and reliable technical expertise to ensure that client systems are efficient, scalable, and future-ready. Their services span across the complete lifecycle of SAP systems, from planning and design to implementation and ongoing support, enabling organizations to streamline operations and enhance productivity.

However, despite advancements in enterprise systems like SAP, there exists a significant gap in areas such as **career guidance, skill development, and workforce readiness**, especially for students and job seekers aspiring to enter such technical domains. Traditional career guidance methods are often manual, generalized, and lack personalization, making it difficult for individuals to align their skills with industry requirements. This creates a need for intelligent, automated solutions that can provide personalized career recommendations and bridge the gap between academic knowledge and industry expectations.

To address this challenge, the consultancy project titled **"CareerForge AI: AI-Based Career and Skill Recommendation System"** was undertaken. The project aims to leverage Artificial Intelligence to provide personalized career guidance, skill gap analysis, resume generation, and interview preparation support. By integrating AI-driven insights with modern web technologies, the system offers a scalable and

efficient platform that complements industry requirements and supports users in making informed career decisions.

Overall, this consultancy initiative aligns with the industry's need for digital transformation and workforce readiness by combining enterprise-level technological understanding with intelligent career guidance solutions, thereby creating value for both users and organizations.

OBJECTIVES OF CONSULTANCY PROJECT

The primary objective of the consultancy project **"AI CareerPilot : AI-Based Career and Skill Recommendation System"** is to design, develop, and deploy an intelligent, scalable, and user-centric platform that delivers personalized career guidance using advanced Artificial Intelligence techniques. The system aims to bridge the critical gap between academic learning and industry requirements by providing data-driven insights and recommendations tailored to individual users based on their skills, interests, qualifications, and career aspirations. By automating the career guidance process, the project seeks to reduce dependency on traditional manual counseling methods, improve decision-making efficiency, and enhance the overall effectiveness and accessibility of career planning services for students and job seekers.

A key objective of the project is to develop a robust AI-driven career recommendation engine capable of analyzing diverse user inputs and mapping them to suitable career paths aligned with current industry trends and opportunities. In addition, the system focuses on implementing a comprehensive skill gap analysis module, which identifies the gap between a user's existing skill set and the competencies required for their desired career, and provides targeted recommendations such as courses, certifications, and learning paths to bridge this gap. Another important objective is to automate the generation of resumes and cover letters using AI, ensuring that users can create professional, well-structured, and Applicant Tracking System (ATS)-friendly documents that significantly improve their chances of success in recruitment processes.

The project also aims to integrate an AI-based mock interview simulation system, enabling users to practice real-world interview scenarios through dynamically generated questions and receive constructive feedback for improvement. Furthermore, the system is designed to provide job and internship recommendations based on user profiles, preferences, and real-time labor market trends, thereby enhancing employability and helping users make informed career choices. Ensuring data security, privacy, and ethical handling of user information is another critical objective, achieved through secure authentication mechanisms, encrypted data storage, and controlled access policies that safeguard sensitive user dat

SCOPE OF WORK

The scope of the consultancy project **"AI CareerPilot : AI-Based Career and Skill Recommendation System"** encompasses the design, development, and deployment of an intelligent platform that provides personalized career guidance using Artificial Intelligence. The project covers both functional and non-functional aspects required to build a scalable, secure, and efficient system that meets the needs of users and client organizations.

Functional Scope

The functional scope of the system includes the development and integration of multiple modules that work together to deliver a comprehensive career guidance solution. The system provides a secure user authentication and login mechanism, enabling users to register, log in, and manage their profiles. A key component of the system is the **AI-based career recommendation engine**, which analyzes user inputs such as skills, interests, and educational background to suggest suitable career paths.

The platform also includes a **skill gap analysis module**, which identifies the difference between the user's current skills and the requirements of selected career paths, and recommends necessary improvements. Additionally, the system supports **automated resume and cover letter generation**, allowing users to create professional, structured, and ATS-friendly documents using AI assistance.

To enhance user preparedness, the system integrates a **mock interview simulation module**, where users can practice interview questions and receive feedback for improvement. The platform further provides a **job and internship recommendation system**, which suggests relevant opportunities based on user profiles and preferences. An **interactive dashboard** is also included, enabling users to track their progress, analyze performance, and monitor career readiness.

Non-Functional Scope

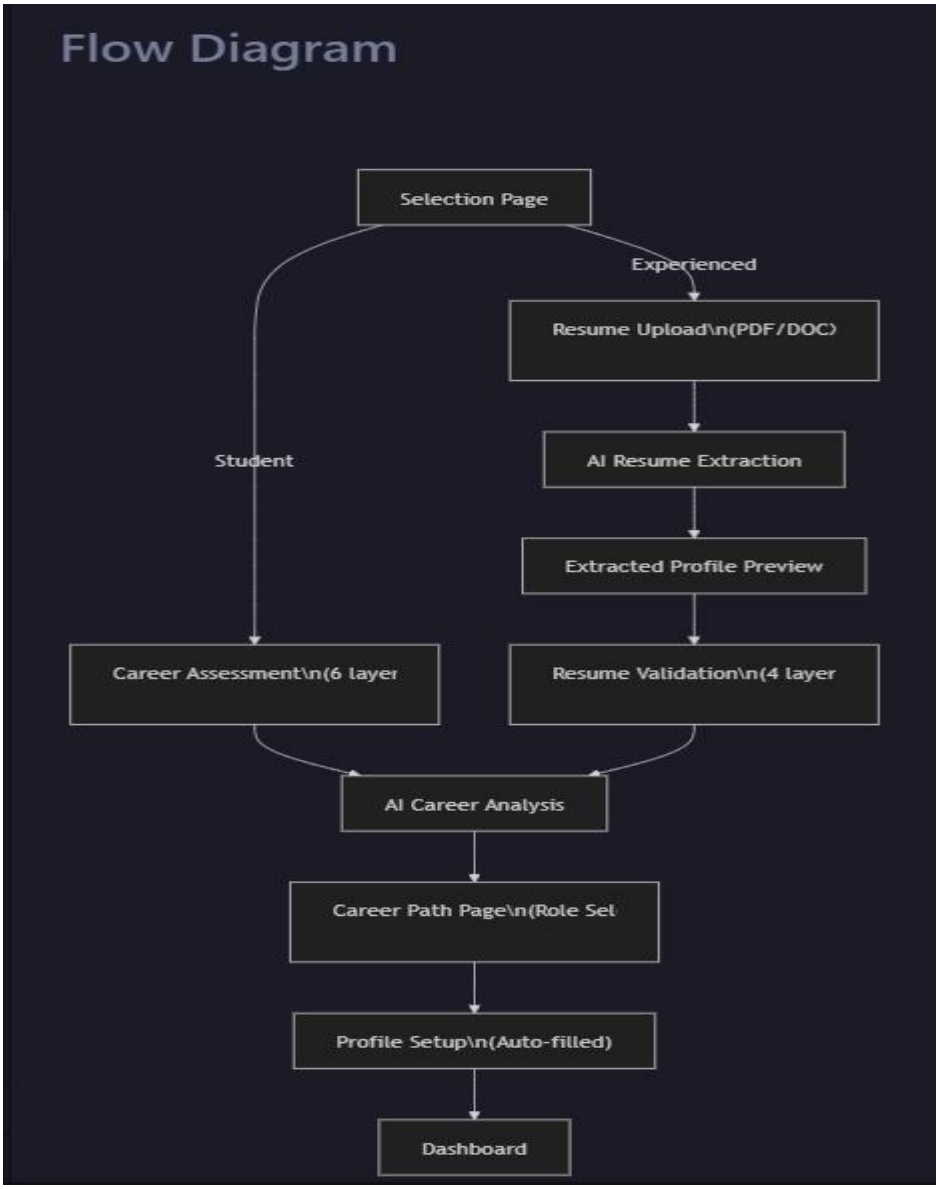
The non-functional scope focuses on ensuring the quality, performance, and reliability of the system. The platform is designed with strong **security measures**, including authentication, authorization, and data protection mechanisms to ensure user privacy. The system is built to be **scalable**, allowing it to handle multiple users simultaneously without performance degradation.

Performance optimization techniques are implemented to ensure **fast response times and real-time processing**, particularly for AI-based recommendations. The system also emphasizes **data privacy and integrity**, ensuring that sensitive user information is securely stored and managed. Additionally, the platform is designed to be **user-friendly and accessible**, providing an intuitive interface and smooth navigation for users with varying levels of technical expertise.

Deployment Scope

The scope also includes the **deployment of the system on a cloud-based environment**, ensuring accessibility, scalability, and reliability. The deployed system supports real-time usage and can be easily updated or extended in the future. Necessary documentation, user guidance, and basic training support are also considered within the scope to ensure effective utilization of the system by the client.

The scope of work further extends to testing and validation, where the system is evaluated for functionality, performance, and usability. The final deliverables include a fully functional AI-driven platform, documentation, and demonstration of system capabilities to the client organization.


Flow Diagram
Selection Page
Experienced
Student
Resume Upload\n(PDF/DOC)
AI Resume Extraction
Extracted Profile Preview
Career Assessment\n(6 layer
Resume Validation\n(4 layer
AI Career Analysis
Career Path Page\n(Role Sel
Profile Setup\n(Auto-filled)
Dashboard

PROPOSED METHODOLOGY

The development of the consultancy project **"AI CareerPilot: AI-Based Career and Skill Recommendation System"** followed a structured and systematic methodology to ensure efficient execution, high-quality output, and successful delivery. The methodology consists of multiple phases, starting from requirement analysis to final deployment, with continuous evaluation and improvement at each stage.

7.1 Requirement Analysis

The project began with a detailed requirement analysis phase, where discussions were conducted with stakeholders to understand the challenges in traditional career guidance systems and identify key system requirements. Functional requirements such as career recommendation, skill gap analysis, resume generation, interview simulation, and job recommendations were identified. Non-functional requirements including scalability, performance, and data security were also documented. This phase ensured a clear understanding of the project scope and objectives.

7.2 System Design

Based on the gathered requirements, a well-defined system architecture was designed. The design phase included the development of a modular structure consisting of frontend, backend, and AI components. The database schema was carefully designed to efficiently manage user data, career profiles, and system interactions. User workflows, data flow diagrams, and module interactions were defined to ensure smooth system functionality and seamless user experience. The design also incorporated security and scalability considerations.

7.3 Development

In the development phase, the system was implemented using modern technologies such as Next.js for frontend development, Node.js for backend services, and PostgreSQL for database management. Artificial Intelligence techniques, including Natural Language Processing (NLP) and Large Language Models (LLMs), were integrated to enable intelligent career recommendations, resume generation, and

interview simulation. Different modules of the system, such as authentication, recommendation engine, skill analysis, and dashboard, were developed and integrated systematically to ensure cohesive functionality.

7.4 Testing

The developed system underwent rigorous testing to ensure functionality, performance, and reliability. Unit testing was performed to validate individual components, followed by integration testing to ensure smooth interaction between modules. User acceptance testing (UAT) was conducted to verify that the system meets user requirements and expectations. Identified issues and bugs were resolved through iterative improvements, enhancing system stability and accuracy.

7.5 Deployment

After successful testing, the system was deployed on a cloud-based platform to ensure accessibility, scalability, and real-time performance. The deployment process included server configuration, database setup, and integration of all system components. Basic user training and documentation were provided to ensure effective utilization of the system. The deployed solution is designed to support future enhancements and easy integration with external platforms.

The final stage of the methodology involves job and internship recommendation. The system integrates external APIs to fetch real-time job opportunities and filters them based on user preferences, skills, and location. To optimize performance, a caching mechanism is implemented to store frequently accessed data, reducing latency and minimizing external API calls. The entire system operates on a serverless architecture, ensuring scalability, reliability, and efficient resource utilization.

Overall, the proposed methodology establishes a continuous feedback loop where user inputs are analyzed, processed, and refined through AI-driven modules, resulting in improved outputs and enhanced user experience. This integrated approach ensures that the system not only provides recommendations but also supports users throughout their career development journey.

TECHNOLOGY STACK USED

The successful implementation of the **AI-Powered Career and Skill Recommendation System** relies on a carefully selected and modern technology stack that ensures scalability, performance, security, and seamless integration of Artificial Intelligence capabilities. The chosen technologies are aligned with industry standards and are specifically selected to support the development of a full-stack, AI-driven web application capable of handling real-time data processing and user interactions efficiently.

The frontend of the system is developed using advanced web frameworks that enable the creation of dynamic, responsive, and user-friendly interfaces. These technologies ensure that users can interact seamlessly with the system across various devices, providing a smooth experience during tasks such as data input, resume generation, and interview simulation. The use of component-based architecture allows for modular development, making the system easy to maintain and extend.

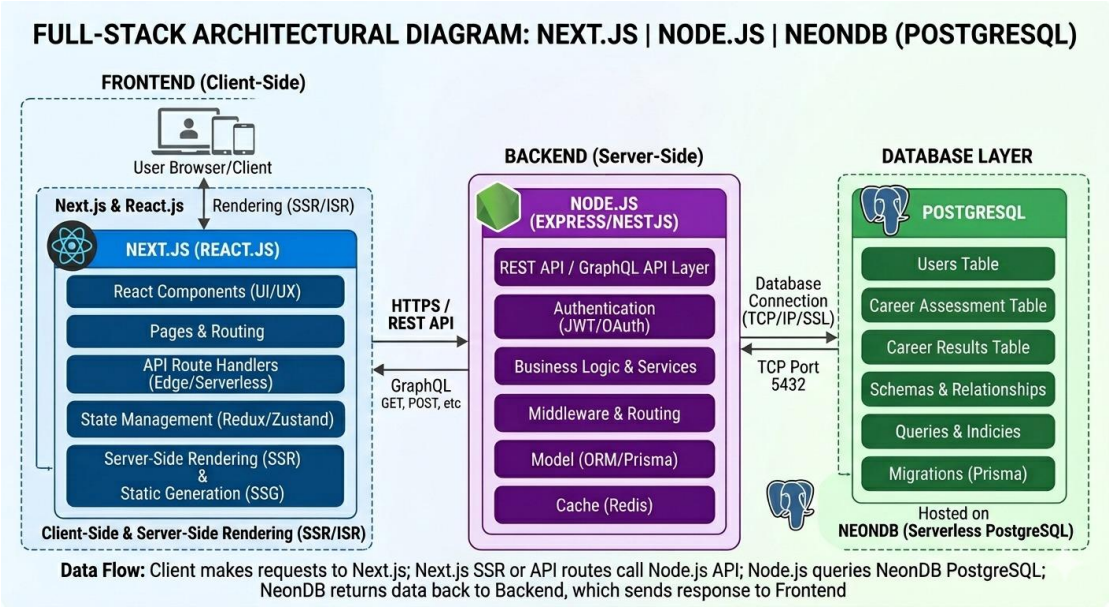
The backend is implemented using a robust runtime environment that efficiently handles server-side logic, API communication, and data processing. It plays a crucial role in managing user requests, executing business logic, and integrating Artificial Intelligence services. The backend also ensures secure handling of sensitive data and maintains communication between the frontend, database, and external APIs. The use of server-side processing enhances system performance and prevents exposure of critical system components.

A key aspect of the technology stack is the integration of Artificial Intelligence through Large Language Models (LLMs). These models are responsible for performing complex tasks such as career assessment, resume generation, and interview evaluation. By leveraging AI, the system is able to understand user inputs contextually and generate personalized outputs, significantly improving the accuracy and effectiveness of the platform.

The data layer is implemented using a relational database system that provides structured storage for user profiles, assessment results, resumes, and job data. An Object-Relational Mapping (ORM) tool is utilized to simplify database interactions

and ensure type safety, making the system more reliable and easier to manage. Additionally, caching mechanisms are incorporated to optimize performance by reducing redundant data retrieval and minimizing latency.

To ensure scalability and cost efficiency, the system is deployed using a serverless architecture. This allows the application to automatically scale based on user demand while minimizing infrastructure management overhead. External APIs are integrated to provide real-time job and internship data, further enhancing the system's functionality and relevance. Overall, the selected technology stack provides a strong foundation for building a high-performance, secure, and intelligent career guidance platform.



SYSTEM ARCHITECTURE

The system architecture of the **AI-Powered Career and Skill Recommendation System** is designed to provide a scalable, secure, and high-performance platform capable of handling complex AI-driven operations. The architecture follows a modern multi-tier design approach, integrating frontend, backend, artificial intelligence services, and data storage into a cohesive and efficient system. This layered structure ensures modularity, allowing each component to operate independently while maintaining seamless communication with other modules.

At the presentation layer, the system provides a user-friendly interface through which users can interact with various functionalities such as career assessment, resume generation, mock interviews, and job search. This layer is responsible for capturing user inputs and displaying processed outputs in an intuitive format. The design ensures responsiveness and accessibility across different devices, enhancing user experience and engagement.

The application layer acts as the core processing unit of the system. It handles business logic, user authentication, request processing, and communication with external services. When a user submits data, the application layer validates the input and forwards it to the appropriate modules for further processing. It also manages interactions with the Artificial Intelligence engine, ensuring that requests are securely transmitted and responses are properly formatted before being returned to the user.

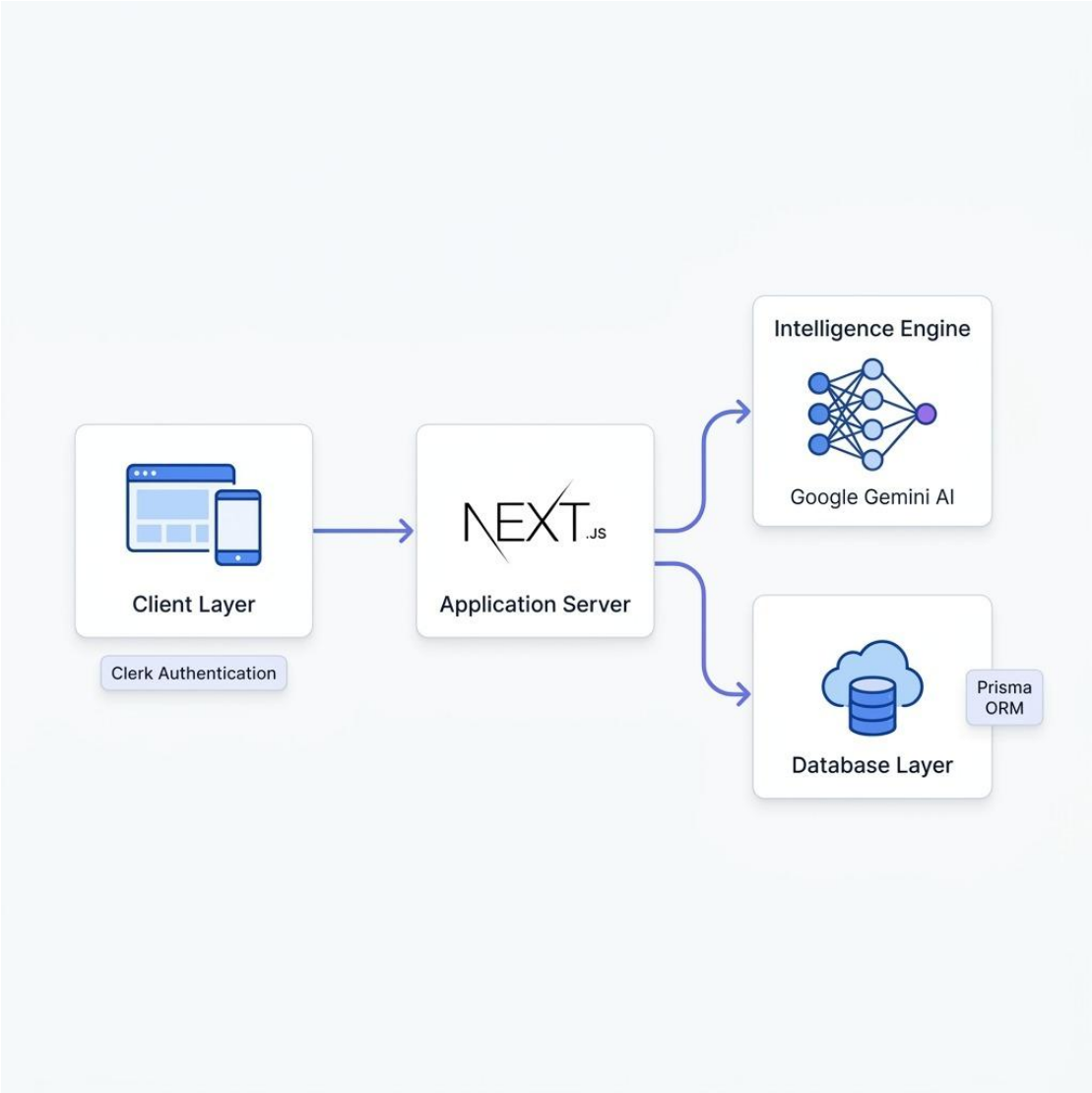
A key component of the architecture is the AI integration layer, which utilizes Large Language Models (LLMs) to perform advanced operations such as career recommendation, resume optimization, and interview evaluation. This layer processes user data using natural language understanding and generates personalized outputs. The integration is designed to be efficient and secure, ensuring that sensitive data is protected while maintaining high processing speed.

The data layer is responsible for storing and managing all system data, including user profiles, assessment results, resumes, and job listings. A relational database system is used to ensure structured storage and efficient retrieval of information. An Object-Relational Mapping (ORM) tool facilitates seamless interaction between the

application and the database, ensuring data integrity and reducing development complexity.

Additionally, the system incorporates an external services layer, which integrates third-party APIs for job and internship data. To enhance performance, a caching mechanism is implemented to store frequently accessed data and reduce redundant API calls. The entire system is deployed using a serverless architecture, allowing automatic scaling based on demand and minimizing infrastructure management.

Overall, the architecture ensures high reliability, efficient resource utilization, and the ability to handle large volumes of users, making it suitable for real-world deployment in an industry environment.



KEY FEATURES DEVELOPED

The consultancy project **"AI CareerPilot: AI-Based Career and Skill Recommendation System"** incorporates a comprehensive set of intelligent and user-centric features aimed at transforming traditional career guidance into a smart, automated, and data-driven process. These features are designed to assist users in making informed career decisions, improving employability, and enhancing overall career readiness by leveraging the power of Artificial Intelligence and modern web technologies.

One of the primary features of the system is the **AI-Based Career Recommendation Engine**, which analyzes user inputs such as skills, interests, educational background, and career preferences to suggest suitable career paths. This engine uses advanced AI techniques to generate personalized recommendations aligned with current industry trends. Supporting this is the **Skill Gap Analysis Module**, which evaluates the difference between the user's existing skill set and the required competencies for their desired career path. Based on this analysis, the system provides actionable suggestions, including recommended skills, courses, and certifications to help users bridge the gap effectively.

Another significant feature is the **Automated Resume and Cover Letter Generator**, which utilizes AI to create professional, structured, and Applicant Tracking System (ATS)-friendly documents. This feature minimizes manual effort while improving the quality and effectiveness of job applications. The system also includes a **Mock Interview Simulation Module**, where users can engage in simulated interview scenarios with AI-generated questions. The module provides feedback on responses, helping users improve their communication skills, confidence, and interview performance.

In addition, the platform offers a **Job and Internship Recommendation System**, which suggests relevant opportunities based on user profiles, preferences, and real-time market demands. The **Personalized User Dashboard** serves as a central interface where users can track their progress, view recommendations, manage their profiles, and monitor their overall career development journey.

DELIVERABLES TO CLIENT

As part of the consultancy engagement for the **AI-Powered Career and Skill Recommendation System**, a comprehensive set of deliverables was provided to the client organization to ensure successful implementation, usability, and long-term value. These deliverables were carefully designed to address the organization's recruitment challenges while enabling seamless adoption of the AI-driven platform into their existing workflows.

The primary deliverable is the fully functional AI-Powered Career and Skill Recommendation System web application, developed using modern full-stack technologies and integrated with advanced Artificial Intelligence capabilities. This application provides an end-to-end solution for career assessment, resume generation, mock interview simulation, and job recommendation. The system is designed with a user-friendly interface, ensuring that both candidates and organizational stakeholders can easily interact with its features without requiring extensive technical expertise.

In addition to the core application, detailed technical documentation was delivered to the client. This includes system architecture descriptions, database design, API integration details, and deployment guidelines. The documentation serves as a reference for future maintenance, scalability, and potential enhancements. It also ensures that the client's technical team can independently manage and extend the system as needed.

Another critical deliverable is the AI-based resume and career assessment engine, which enables automated evaluation of candidate profiles and generation of ATS-optimized resumes. This significantly reduces the manual effort involved in screening candidates and improves the quality of recruitment outcomes. The mock interview module, also delivered as part of the system, provides an interactive platform for candidates to practice interviews and receive real-time feedback, thereby enhancing their preparedness.

The project also includes deployment and hosting of the application on a serverless platform, ensuring scalability, high availability, and optimized performance. Along with deployment, the client was provided with configuration details and access

credentials, enabling smooth operation and monitoring of the system. Furthermore, training and demonstration sessions were conducted to familiarize the client's team with the system functionalities and ensure effective utilization.

Finally, performance evaluation reports and testing results were delivered to validate the reliability and efficiency of the system. These reports highlight improvements in recruitment processes, reduced processing time, and enhanced user experience. Collectively, these deliverables ensure that the client organization can effectively leverage the AI Career

Deliverable	**Description**
AI Career Pilot Application	Fully functional AI-based web platform
Technical Documentation	Architecture, database, and API details
Resume Generation Module	ATS-optimized resume creation
Career Assessment Engine	AI-based skill and career analysis
Mock Interview System	Interactive interview simulation
Job Recommendation Module	Real-time job and internship suggestions
Deployment Setup	Serverless hosting and configuration
Training & Support	User training and system demonstration
Testing Reports	Performance and validation results

OUTCOMES AND IMPACT

The implementation of the **AI-Powered Career and Skill Recommendation System** has resulted in significant positive outcomes for the client organization, both in terms of operational efficiency and strategic value. As a consultancy solution, the system successfully addresses the key challenges associated with traditional recruitment and career guidance processes by introducing automation, intelligence, and integration into a single platform.

One of the most notable outcomes is the substantial improvement in recruitment efficiency. The AI-driven career assessment and resume generation modules have automated the initial stages of candidate evaluation, reducing the dependency on manual screening processes. This has enabled the organization to process a larger number of applications in a shorter time frame while maintaining high accuracy in candidate selection. The use of ATS-optimized resumes has further improved the chances of identifying suitable candidates, thereby enhancing the overall quality of recruitment.

The system has also significantly enhanced the candidate experience. By providing personalized career recommendations, structured learning roadmaps, and AI-based mock interviews, the platform empowers users to better prepare for job opportunities. This not only increases candidate confidence but also ensures that applicants are more aligned with the expectations of the organization. The interactive and user-friendly interface further contributes to a seamless and engaging experience.

From a technical perspective, the adoption of a serverless architecture and caching mechanisms has led to improved system performance and scalability. The application is capable of handling multiple concurrent users with minimal latency, ensuring consistent performance even under high load conditions. The integration of external APIs for job recommendations provides real-time and relevant opportunities, making the platform highly dynamic and responsive to market trends.

Another critical impact is the reduction in operational costs. By automating repetitive tasks such as resume screening and interview preparation, the organization can allocate resources more efficiently and focus on strategic activities. Additionally, the

centralized nature of the system eliminates the need for multiple disconnected tools, further reducing maintenance and operational overhead.

From a consultancy standpoint, the project has delivered measurable value by transforming the organization's recruitment process into a data-driven and intelligent system. The AI-Powered Career and Skill Recommendation platform not only improves current operations but also provides a scalable foundation for future enhancements. It enables the organization to stay competitive in a rapidly evolving job market by leveraging advanced technologies and innovative solutions.

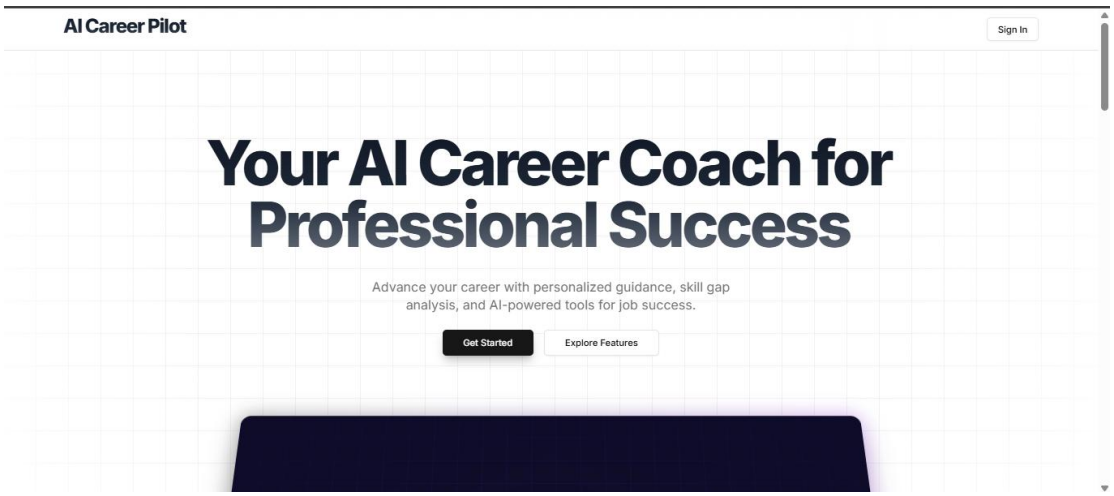


Figure 1: Home Page Interface showing the AI-powered career guidance platform with user-friendly navigation.

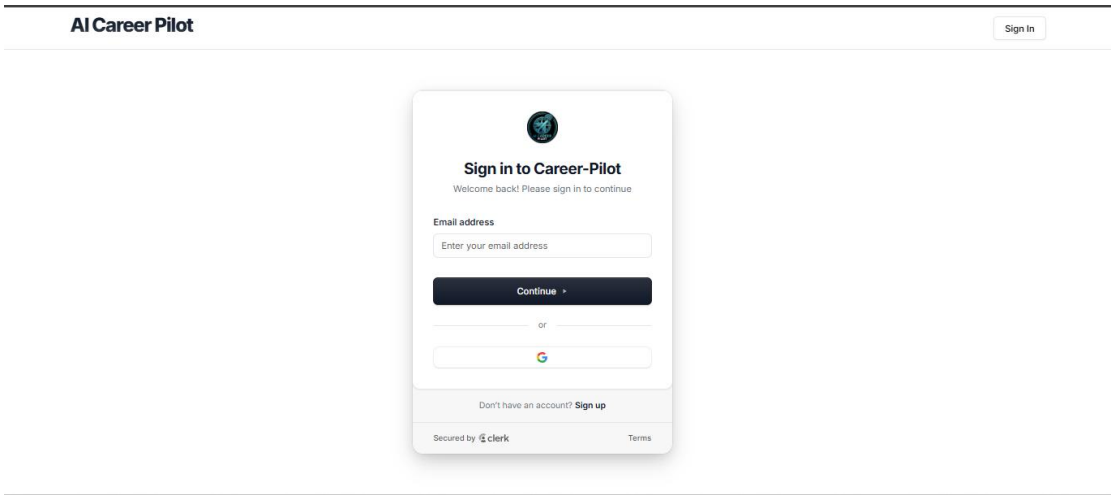


Figure 2: User Authentication Interface enabling secure login and access control.

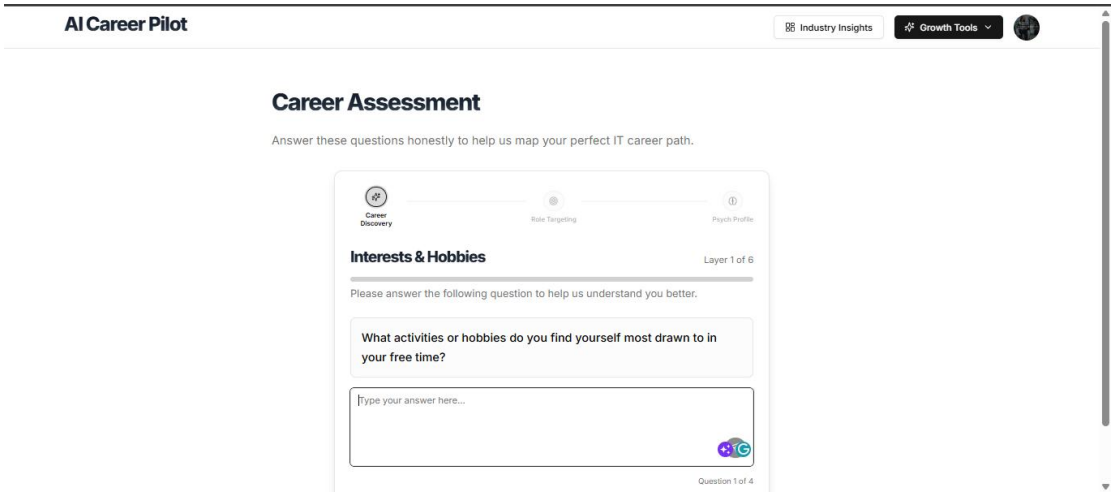
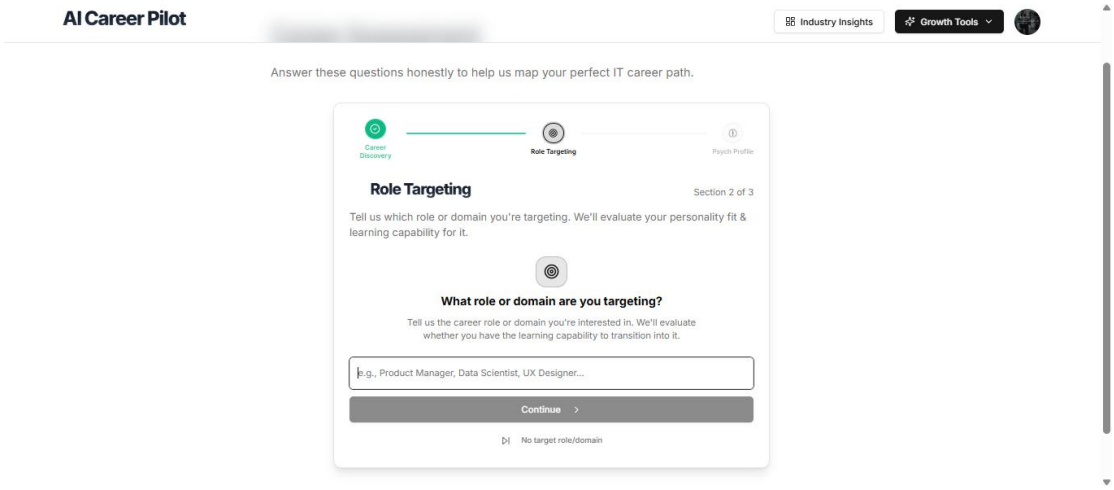


Figure 3: AI-Based Career Recommendation Interface for the career assessment of the user.



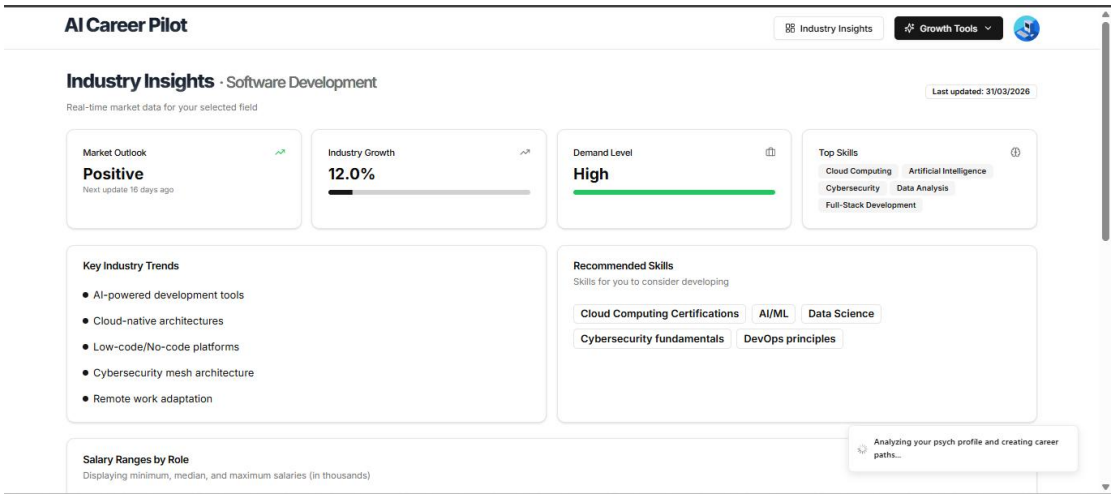


Figure 4: AI-Based Career Recommendation Interface providing personalized career suggestions.

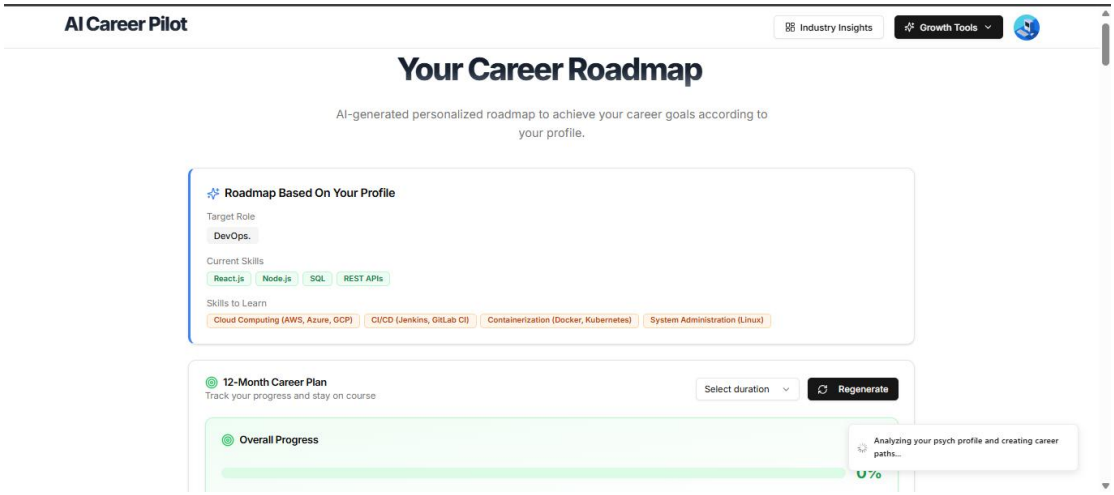


Figure 5: Career Roadmap Interface illustrating a structured, step-by-step progression path with recommended skills and milestones for achieving the selected career goal.

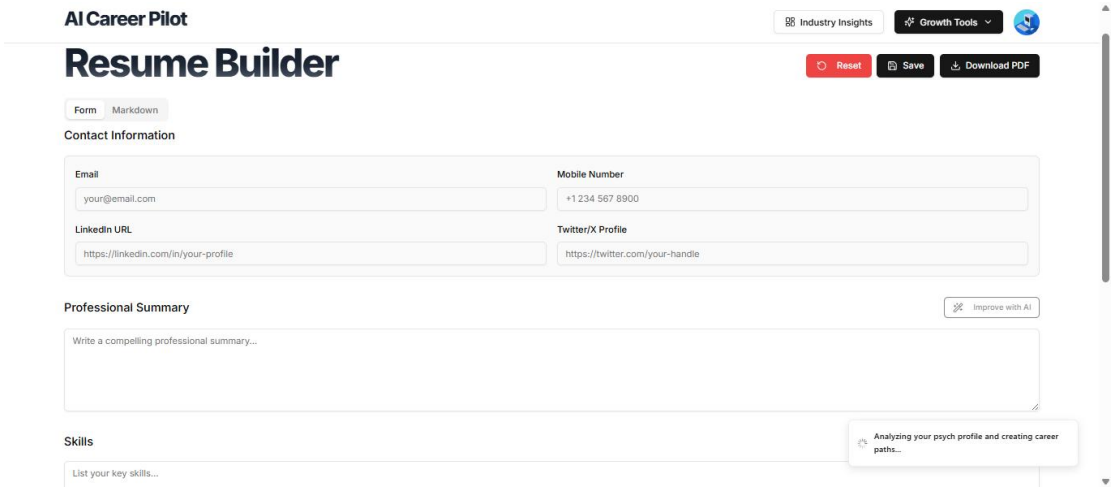


Figure 6: Automated Resume Generation Module creating professional and ATS-friendly resumes.

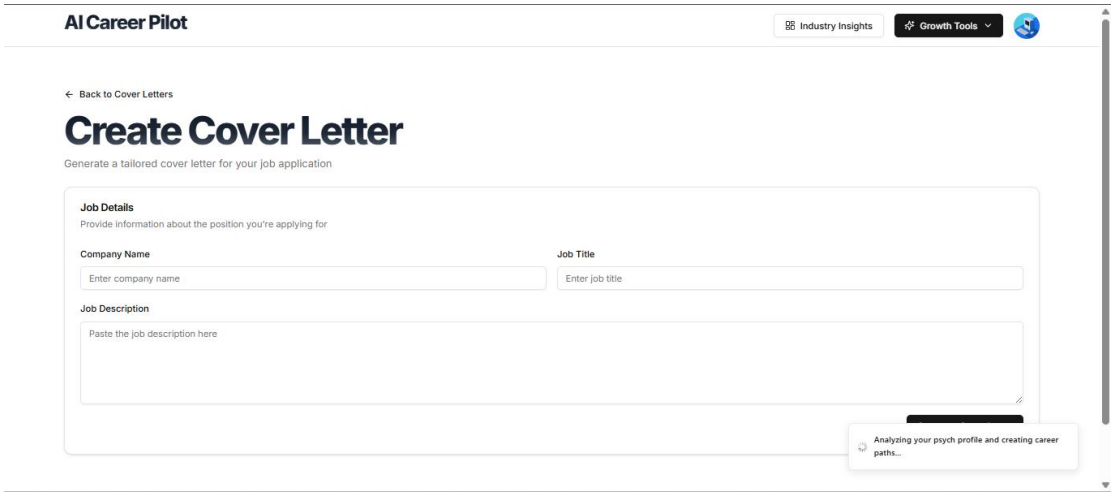
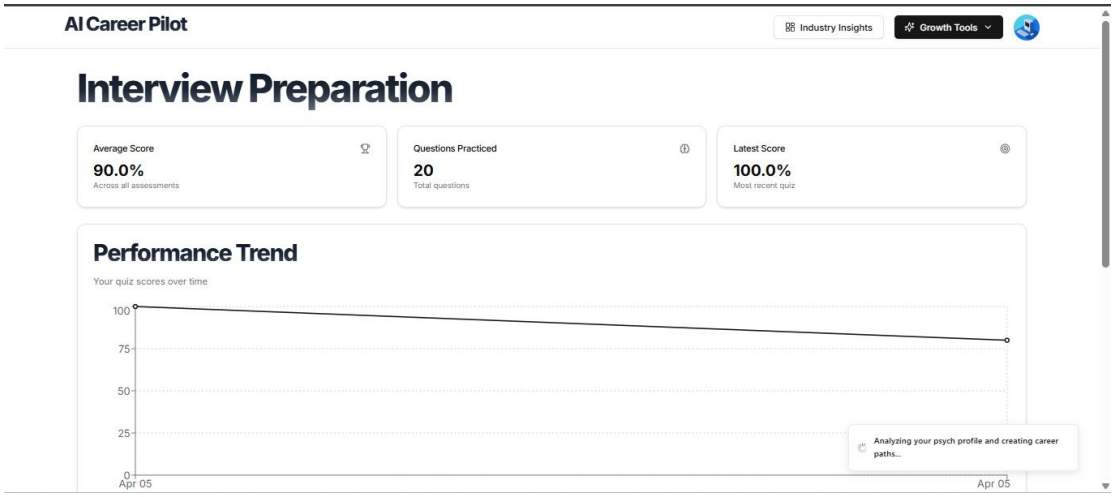


Figure 7: Automated Cover Letter Generation Interface showcasing AI-generated, professional and customized cover letters based on user profile and job requirements.



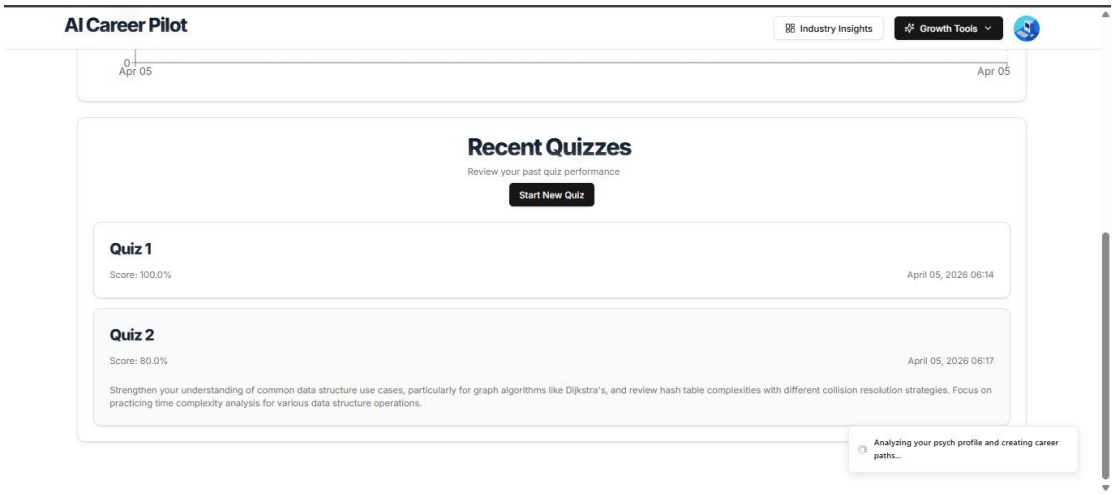


Figure 8 & 9: Mock Interview Simulation Interface for practicing real-world interview scenarios.

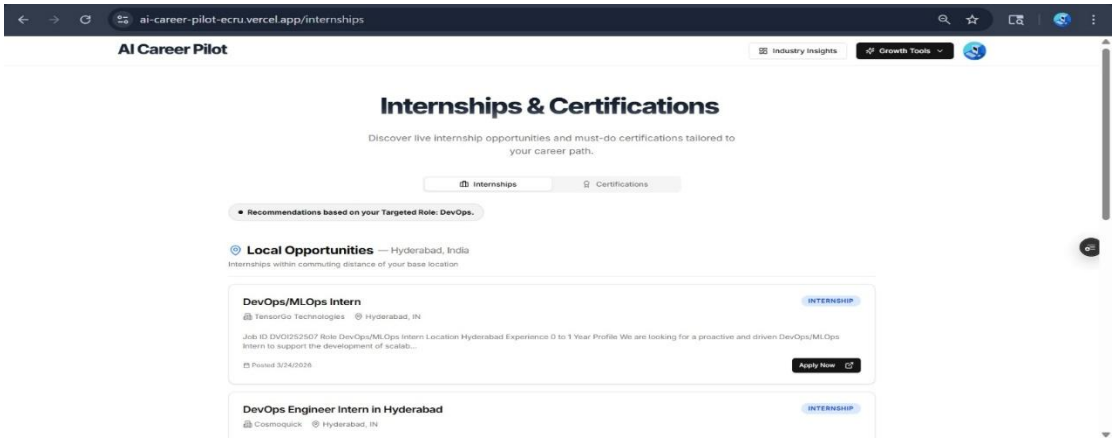


Figure 10: Job and Internship Recommendation System based on user profile and preferences.

REVENUE GENERATED

The consultancy project **"AI CareerPilot: AI-Based Career and Skill Recommendation System"** contributed to institutional revenue through the successful design, development, and deployment of an AI-driven career guidance platform for the client organization. The total consultancy amount generated from this project is estimated to be **₹50,000**, which reflects the value of the technical expertise, development effort, system implementation, and support provided throughout the project lifecycle. This revenue highlights the practical applicability of the project and its alignment with real-world industry requirements.

The revenue generated from the consultancy activity is distributed among the institution, department, and faculty members in accordance with the standard policies and norms followed by the institution. A significant portion of the revenue is allocated to the institution to support infrastructure development, research initiatives, and the promotion of further consultancy and innovation activities. The department receives a share of the revenue to enhance laboratory facilities, support academic programs, and encourage student-centric development initiatives. The remaining portion is distributed among the faculty members involved in the project as recognition of their technical guidance, supervision, and contribution to the successful completion of the consultancy work.

As per the typical distribution model, the revenue is divided as follows: **60% to the institution (₹30,000), 20% to the department (₹10,000), and 20% to the faculty members (₹10,000)**. This structured allocation ensures a balanced and transparent distribution of funds, while also motivating faculty and departments to actively engage in consultancy projects. Furthermore, it promotes a culture of industry collaboration and innovation within the institution.

Overall, the consultancy project not only demonstrates the successful application of Artificial Intelligence in solving real-world problems but also contributes to revenue generation and strengthens the institution's engagement with industry. It aligns with NBA criteria by showcasing measurable financial outcomes, faculty involvement, and the impact of consultancy activities on institutional growth and development.

FACULTY INVOLVEMENT

The successful execution of the consultancy project **"AI CareerPilot : AI-Based Career and Skill Recommendation System"** was made possible through the active involvement, guidance, and technical expertise of the faculty members. Their contributions played a crucial role in ensuring the quality, reliability, and successful completion of the project. The faculty team provided continuous support throughout all phases of the project, including requirement analysis, system design, development, testing, and deployment.

Dr. Raziuddin, serving as the **Project Coordinator**, was responsible for overall project supervision, system design guidance, and strategic decision-making. He ensured that the project objectives were aligned with industry requirements and academic standards. His continuous monitoring and feedback helped maintain the quality and direction of the project, ensuring timely completion and effective implementation.

Dr. Mohammed Abdul Khaleel contributed as the **Technical Advisor**, providing expertise in system architecture, implementation strategies, and technical problem-solving. He played a key role in guiding the development process, ensuring that best practices were followed in coding, system integration, and performance optimization. His involvement also included reviewing the system design, assisting in troubleshooting technical challenges, and supervising testing activities to ensure system reliability.

Dr. Khaja Mizbahuddin Quadry served as the **ML Expert**, focusing on the design, development, and optimization of AI models used in the system. His contributions included guiding the implementation of Natural Language Processing (NLP) techniques, improving the accuracy of career recommendations, and ensuring that AI components function efficiently. He also assisted in refining model outputs and validating the effectiveness of AI-based features.

The collaborative efforts of the faculty team ensured that the project was executed in a structured and professional manner. Their involvement not only enhanced the technical quality of the system but also provided valuable learning and practical

exposure to the student team. The faculty contributions significantly contributed to the successful delivery of a scalable, efficient, and industry-relevant solution.

Faculty Contribution Summary

Name	Role	Contribution
Dr. Raziuddin	Coordinator	System Design, Project Guidance & Review
Dr. Mohammed Abdul Khaleel	Technical Advisor	System Architecture, Implementation Support & Testing Supervision
Dr. Khaja Mizbahuddin Quadry	ML Expert	AI Model Design, Optimization & Validation

STUDENT INVOLVEMENT

The consultancy project was carried out by a team of undergraduate students from the Department of Computer Science & Engineering.

Team Members:

Mohammed Fawzaan – Full Stack Developer

Syed Towhid Hussain Magrabi – Backend Developer

Abu Sufyan – UI/UX Design

Syed Hasnain Tariq – Frontend Developer

The students were actively involved in all phases of the **AI-Powered Career and Skill Recommendation System** consultancy project. Their participation included requirement analysis, system design, development, testing, and deployment of the application in alignment with the client organization's needs.

The team collaboratively worked on implementing core modules such as career assessment, resume generation, mock interview simulation, and job recommendation using modern technologies and Artificial Intelligence techniques. Students also engaged in regular interactions with the industry mentor to understand real-world challenges and incorporate practical solutions into the system.

This hands-on involvement enabled students to gain valuable experience in full-stack development, AI integration, and consultancy practices, thereby enhancing their technical, problem-solving, and professional skills.

CHALLENGES FACED & SOLUTIONS IMPLEMENTED

During the development and deployment of the **AI-Powered Career and Skill Recommendation System**, several technical and operational challenges were encountered. These challenges were primarily associated with integrating advanced Artificial Intelligence models, ensuring system performance, maintaining data security, and aligning the solution with real-world industry requirements. Addressing these challenges required a combination of innovative problem-solving approaches, careful system design, and the adoption of modern development practices.

One of the major challenges faced was the integration of Large Language Models (LLMs) for performing complex tasks such as career assessment, resume generation, and interview evaluation. These models require carefully structured inputs to generate accurate and meaningful outputs. To overcome this, a prompt engineering strategy was implemented, ensuring that user data was formatted and processed effectively before being sent to the AI model. This significantly improved the relevance and quality of generated results.

Another critical challenge was managing system performance, particularly when handling multiple user requests and external API calls for job recommendations. Without optimization, the system experienced latency issues and increased response times. To address this, a caching mechanism based on a Time-To-Live (TTL) strategy was introduced. This allowed frequently requested data to be stored temporarily, reducing redundant API calls and improving overall system responsiveness.

Data security and privacy were also significant concerns, as the system handles sensitive user information such as personal details, skills, and career preferences. Ensuring secure data transmission and storage was essential. This challenge was mitigated by implementing secure authentication mechanisms, encrypted communication protocols, and strict access control policies. These measures ensured that user data remained protected throughout the system.

Another challenge involved designing a system that could cater to both freshers and experienced professionals. Each group has different requirements and expectations, making it difficult to create a one-size-fits-all solution. This was resolved by

developing a dynamic assessment module that adapts its logic based on the user's experience level. This approach ensures that the system provides relevant and personalized recommendations for different user categories.

Integration of multiple technologies, including frontend frameworks, backend services, databases, and external APIs, also posed challenges in terms of compatibility and data flow management. To overcome this, a modular architecture was adopted, allowing each component to function independently while maintaining seamless communication through well-defined interfaces. This improved system reliability and simplified debugging and maintenance.

Overall, the challenges encountered during the project provided valuable insights into real-world system development and contributed to the creation of a robust, efficient, and scalable solution. The implemented solutions not only resolved the immediate issues but also enhanced the overall quality and performance of the system.

SOLUTIONS IMPLEMENTED

During the development of the consultancy project **"AI CareerPilot: AI-Based Career and Skill Recommendation System"**, several technical and operational challenges were encountered. To address these challenges effectively, appropriate solutions were implemented to ensure system performance, reliability, security, and accuracy.

One of the major concerns was ensuring **data security and user privacy**, as the system handles sensitive user information such as personal details, career preferences, and uploaded data. To address this, robust **encryption techniques and secure authentication mechanisms** were implemented. User data is protected using secure communication protocols (HTTPS), password hashing, and controlled access policies to prevent unauthorized access. These measures ensure confidentiality, integrity, and safe storage of user information.

Another significant challenge was achieving **accurate and relevant AI-based recommendations**. Since the system relies on Artificial Intelligence models such as Natural Language Processing (NLP) and Large Language Models (LLMs), initial outputs required optimization to improve quality and relevance. This was addressed through **model optimization techniques**, including prompt engineering, fine-tuning responses, and refining input parameters to generate more precise and personalized recommendations. Continuous testing and iterative improvements were carried out to enhance the overall performance and reliability of the AI modules.

The system also faced challenges related to **performance and response time**, especially while handling real-time AI interactions and multiple user requests. To overcome this, **performance optimization strategies** such as caching frequently generated responses, efficient API handling, and asynchronous processing were implemented. These techniques reduced latency and ensured faster response times, thereby improving the overall user experience.

Additionally, to ensure **scalability and smooth system operation**, the application was deployed using a cloud-based architecture with efficient resource management. Proper database optimization and structured query handling were implemented to

handle large volumes of user data without performance degradation. The system was also designed to support future enhancements and integration with external platforms.

Overall, the implemented solutions successfully addressed key technical challenges related to security, performance, and AI accuracy. These improvements ensured that the system operates efficiently, delivers reliable results, and provides a secure and seamless experience to users, thereby enhancing the overall effectiveness of the consultancy project.

CLIENT FEEDBACK

The client organization provided highly positive feedback regarding the implementation and performance of the **AI-Powered Career and Skill Recommendation System**. As a mid-sized IT and Knowledge Process Outsourcing (KPO) company, the client emphasized the significant improvements achieved in their recruitment and candidate evaluation processes. The system was recognized as a valuable addition to their operational framework, particularly due to its ability to automate repetitive tasks and provide intelligent, data-driven insights.

From the client's perspective, one of the most impactful aspects of the system is its ability to deliver personalized career recommendations and generate ATS-optimized resumes. The organization noted that this feature has considerably enhanced the quality of candidate profiles, making it easier to identify suitable applicants. The mock interview module was also highly appreciated, as it provides candidates with a realistic preparation environment, resulting in better performance during actual interviews. This, in turn, reduces the effort required by recruiters to assess candidate readiness.

The client also highlighted the efficiency and performance of the system. The use of a serverless architecture and caching mechanisms ensures that the application operates smoothly with minimal latency, even when handling multiple users simultaneously. The integration of real-time job recommendation APIs was recognized as a key strength, enabling the system to provide relevant and up-to-date opportunities. Additionally, the user-friendly interface and intuitive design were praised for making the platform accessible to both technical and non-technical users.

Another important aspect of the feedback was related to the system's scalability and future potential. The client acknowledged that the modular architecture allows for easy integration of additional features, such as advanced analytics and voice-based interview systems. The consultancy team's ability to deliver a solution aligned with industry requirements and organizational goals was also commended. Overall, the client expressed satisfaction with the project outcomes and indicated interest in further collaboration for future enhancements.

FUTURE SCOPE

The **AI-Powered Career and Skill Recommendation System** establishes a strong foundation for intelligent career guidance and recruitment automation; however, there exists significant scope for further enhancement and expansion to meet evolving industry demands. As the fields of Artificial Intelligence and recruitment technologies continue to advance, the system can be extended to incorporate more sophisticated features that improve accuracy, interactivity, and overall user experience.

One of the most promising areas for future development is the integration of voice-based and video-based interview analysis. By incorporating speech recognition and computer vision technologies, the system can evaluate not only the content of user responses but also non-verbal communication aspects such as tone, facial expressions, and confidence levels. This would provide a more comprehensive assessment of candidate performance and better simulate real-world interview scenarios.

Another potential enhancement involves the implementation of advanced machine learning models for predictive analytics. By analyzing historical data and user behavior, the system can predict career trends, recommend emerging skill sets, and provide proactive guidance to users. This would enable candidates to stay ahead of industry changes and make informed decisions regarding their career paths. Additionally, the system can be improved by integrating with professional networking platforms, allowing users to synchronize their profiles and apply for jobs directly through the platform.

The scalability of the system can also be further enhanced by incorporating multi-language support and expanding its reach to a global audience. This would make the platform accessible to users from diverse backgrounds and increase its applicability across different regions. Furthermore, the inclusion of personalized learning modules, such as curated courses and certification recommendations, can help users bridge their skill gaps more effectively.

MEMORANDUM OF UNDERSTANDING (MoU)

I. Introduction

This Memorandum of Understanding (MoU) is made and entered into on **1st Jan 2026** between:

Lords Institute of Engineering & Technology, Department of Computer Science Engineering, Hyderabad

AND

Zenith TEKSystems LLP

The purpose of this MoU is to establish a collaborative framework for the development and implementation of the consultancy project titled **"AI CareerPilot: AI-Based Career and Skill Recommendation System"**, aimed at providing an intelligent, AI-driven platform for career guidance and skill developmen

II. Scope of Work and Responsibilities

Both parties agree to collaborate and contribute towards the successful execution of the consultancy project as per the defined scope.

Responsibilities of the Institution:

- Design and development of the AI-based career recommendation system
- Implementation of modules such as skill gap analysis, resume generation, and mock interview simulation
- Testing, validation, and deployment of the system
- Providing technical documentation and demonstration of the system

Responsibilities of the Client:

- Providing necessary inputs and requirements for system development
- Reviewing and validating system functionality
- Providing feedback for improvements and enhancements

- Supporting evaluation and acceptance of deliverables

IV. Deliverables

- Fully functional AI-based career guidance system
- Source code and technical documentation
- User manual and system guidelines
- Demonstration and training (if required)

V. Financial Terms

This project is carried out as part of an **academic consultancy initiative**.
The total consultancy amount is ₹50,000 to be distributed as per institutional norms.

VI. Duration of Agreement

This MoU shall remain valid from 1st Jan 2026 to 30th July 2026, unless extended by mutual agreement of both parties.

VII. Confidentiality

Both parties agree to maintain confidentiality of all project-related information, data, and materials, and shall not disclose any information to third parties without prior consent.

IX. Signatories

This MoU is signed by the authorized representatives of both parties.

Institution	Client Organization
Name: Dr. Syed Raziuddin	**Name:** Eliyas
Designation: Head of Department (CSE)	**Designation:** Managing Director
Signature:	**Signature:**